6–LOADED CENTRE OF GRAVITY

6.1 GENERAL PRINCIPLES

6.1.1 Effect of Adding a Mass to a Body

Consider a homogenous bar of mass M to be initially supported at its centre of gravity (C of G), (G), in a horizontal posture as shown in Figure 6.1.

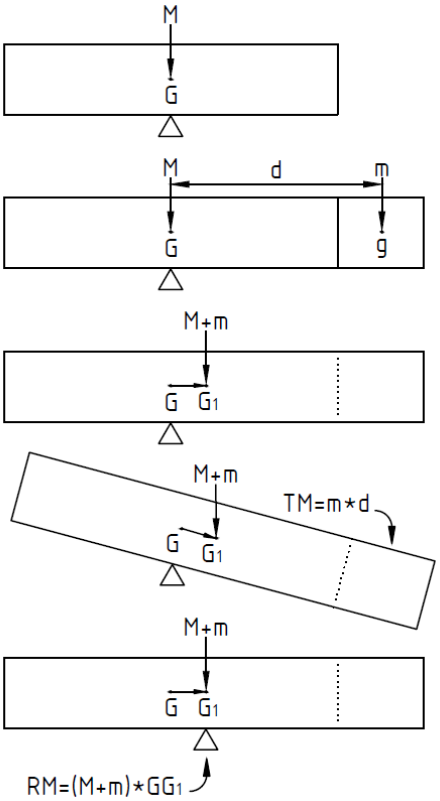


Figure 6.1 – Effect of Adding a Mass

If now a similar homogenous bar with mass (m) and C of G, (g) is attached at a distance (d) as shown, the result will be a new bar of mass (M+m) with C of G at (G_1).

With the new bar still supported at G, a Tilting Moment (TM) will come into action to rotate the bar clockwise.

To restore the new bar to the original horizontal condition, an equal but opposing Righting Moment (RM) will have to be applied at G_1.

Whereby: $\mathrm{RM} = \mathrm{TM}$

$$(\mathrm{M} + \mathrm{m}) * \mathrm{GG}_1 = \mathrm{m} * \mathrm{d}$$

Therefore the shift in C of G:

$$GG_1 = \frac{m*d}{M+m}$$

Summary

When a mass is *added* to a body the C of G of the body *shifts towards* the C of G of the *added mass* by an amount equal to the product of the mass added, and the distance between the C of Gs of the mass and the body, divided by the new total mass.

6.1.2 Effect of Removing a Mass from a Body

Consider this time that the mass is rather removed as shown in Figure 6.2. This will result in a new bar of mass (M-m) with C of G at (G_1).

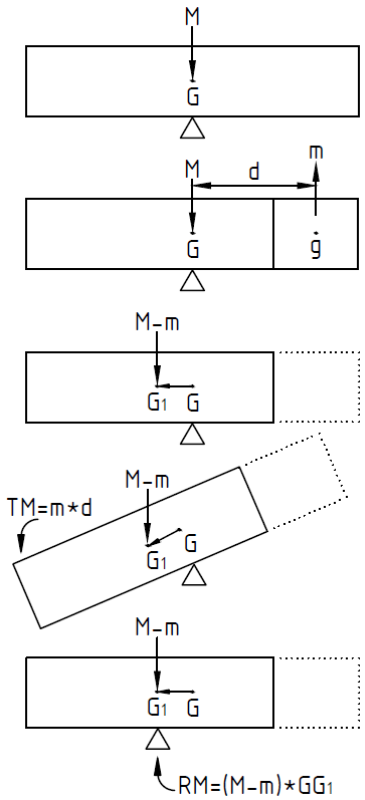


Figure 6.2 – Effect of Removing a Mass

With the new bar still supported at G, a TM will come into action to rotate the bar anticlockwise.

Again to restore the new bar to the original horizontal condition, an equal but opposing RM will have to be applied at G_1.

Now: $$RM = TM$$

$$(M - m) * GG_1 = m * d$$

Therefore the shift in C of G:

$$GG_1 = \frac{m * d}{M - m}$$

Summary

When a mass is *removed* from a body the C of G of the body *shifts away* from the C of G of the *removed mass* by an amount equal to the product of the mass removed and the distance between the C of Gs of the mass and the body, divided by the new total mass.

6.1.3 Effect of Shifting a Mass within a Body

Consider a body of total mass M to be composed of six small masses of sizes (m) each as shown in Figure 6.3.

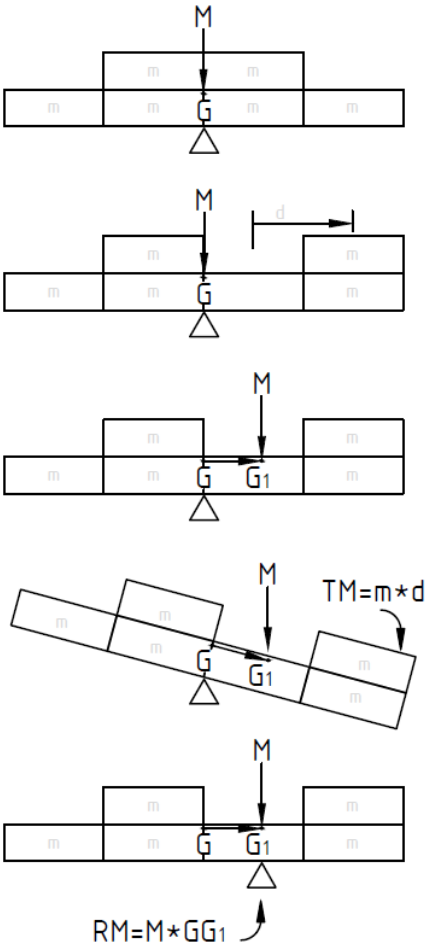


Figure 6.3 – Effect of Shifting a Mass

Now shift one mass to the right as shown. The C of G of the body will immediately shift to G_1.

With the bar still supported at G, a TM will come into action to rotate the bar clockwise.

To restore the body back to the original horizontal condition, an equal but opposing RM will have to be applied at G_1.

Now: $$RM = TM$$

$$M * GG_1 = m * d$$

Therefore the shift in C of G:

$$\mathrm{GG_1} = \frac{\mathrm{m} * \mathrm{d}}{\mathrm{M}}$$

Summary

When a mass is *shifted within* a body the C of G of the body *shifts parallel* to the C of G of the *shifted mass* by an amount equal to the product of the mass shifted and the distance shifted, divided by the total mass.

6.1.4 Effect of Suspending a Mass

In the left diagram of Figure 6.4 a mass (m) is attached by rope to the lifting gear and ready to be hoisted. In this slack condition there is no tension in the rope and the mass may be considered as not forming part of the gear.

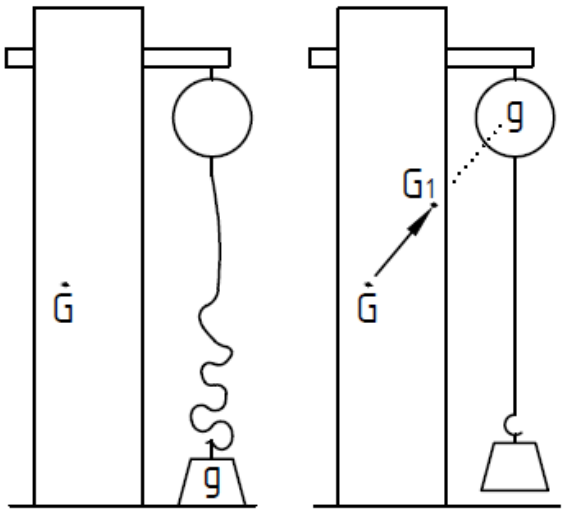


Figure 6.4 – Effect of Suspending a Mass

Immediately the mass is lift off the ground as in the right diagram the rope becomes tensioned and pressure is put on the hook at the top of the gear.

Summary

When a mass is *suspended* the C of G of the mass *shifts* to the point of suspension, and no matter how far away the mass is from the point of suspension, the effect remains the same.

6.2 POSITION OF CENTRE OF GRAVITY OF A SHIP

The centre of gravity of a ship, being the centroid of a three-dimensional body is described by three coordinates. These are:

6.2.1 The **Vertical Centre of Gravity** (VCG_K)

This position is measured with respect to the keel. It is usually written as KG.

6.2.2 The **Longitudinal Centre of Gravity** (LCG_{MS})

It is measured with respect to midship. Its fore or aft direction hence has to be specified in every particular case.

6.2.3 The **Transverse Centre of Gravity** (TCG_{CL})

Measured with respect to the centreline, the TCG_{CL} will usually have a zero value since ships are normally constructed with symmetrical hull forms and balanced distribution of machinery and equipment port–starboard.

6.3 LIGHTSHIP CENTRE OF GRAVITY

The Longitudinal position in the lightship condition (LCG_{MSL}) is at the same distance from midship as LCB_L as shown in Figure 6.5(a) and need not further be calculated.

In respect of the vertical position KG_L, Figure 6.5(b), this is determined by conducting an experiment on a ship when construction is completed, and unless a ship is modified, the position remains constant throughout its service life. The method of conducting the experiment will be discussed later in Chapter 10.

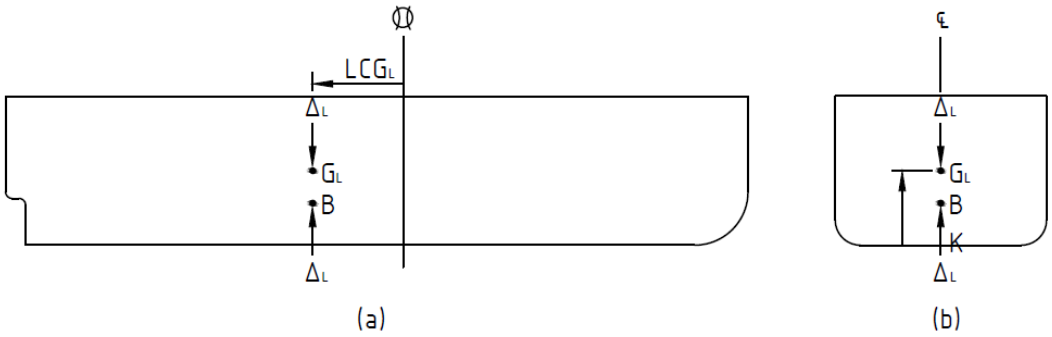


Figure 6.5 – Lightship Centre of Gravity

The transverse position usually coincides with the centreline and will therefore have a value of zero.

6.4 LOADED CENTRE OF GRAVITY

In any condition of loading over and above the lightship, the three positions of the centre of gravity are determined by taking moments as follows:

6.4.1 Determination considering added and removed masses together first followed by computation of shifted masses

6.4.1.1 Determination of KG

Consider a vessel initially empty with lightship displacement Δ_L to have its C of G at G_L and hence a lightship C of G (KG_L) as in Figure 6.5(b).

Now add masses m_1, m_2, m_4 and m_6 at heights z_1, z_2, z_4 and z_6 respectively, and discharge masses m_3 and m_5 from heights z_3 and z_5 respectively.

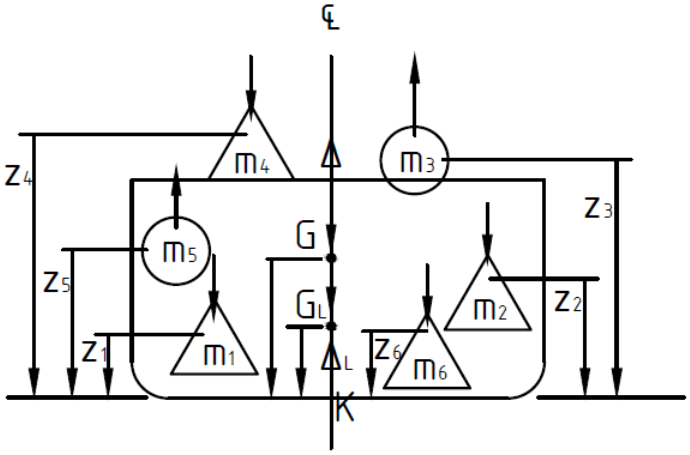


Figure 6.6 – Loaded KG

Its new total mass will be: $\Delta = \Delta_L + m_1 + m_2 + m_4 + m_6 - m_3 - m_5$

With a new C of G (KG). By moments of masses about the keel:

$$\text{Final moment} = \text{Sum of moments of individual masses}$$

$$\Delta KG = \Delta_L KG_L + m_1 z_1 + m_2 z_2 + m_4 z_4 + m_6 z_6 - m_3 z_3 - m_6 z_5$$

$$KG = \frac{\Delta_L KG_L + m_1 z_1 + m_2 z_2 + m_4 z_4 + m_6 z_6 - m_3 z_3 - m_6 z_5}{\Delta}$$

For ease of computation and to reduce making mistakes the above calculations may be presented vertically as follows:

Masses	Kg	Moments
Δ_L	KG_L	$\Delta_L KG_L$
m_1	z_1	$m_1 z_1$
m_2	z_2	$m_2 z_2$
$-m_3$	z_3	$-m_3 z_3$
m_4	z_4	$m_4 z_4$
$-m_5$	z_5	$-m_5 z_5$
m_6	z_6	$m_6 z_6$
$\Delta=\sum$(Masses)		$\sum$(Moments)

Table 6.1 – Calculation for KG

Therefore $$KG = \frac{\sum(\text{Moments})}{\sum(\text{Masses})}$$

6.4.1.2 Determination of LCG_{MS}

If now moments are taken with reference to midship (MS), then by the vertical computation method, and considering aft moments to be positive and forward moments to be negative:

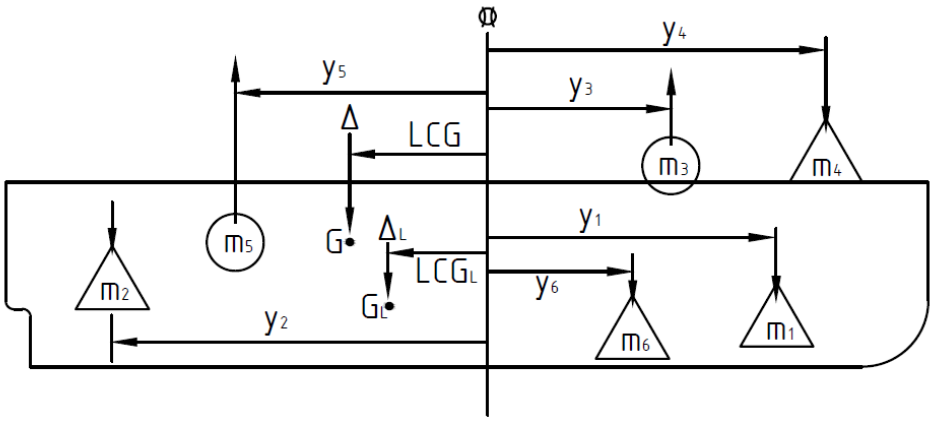


Figure 6.7 – Loaded LCG

Masses	Lcg_{MS}	Moments	
		Aft (+)	Fwd (-)
Δ_L	$LCG_{LA}A$	$\Delta_L LCG_L$	
m_1	y_1F		m_1y_1
m_2	y_2A	m_2y_2	
$-m_3$	y_3F		$-m_3y_3$
m_4	y_4F		m_4y_4
$-m_5$	y_5A	$-m_5y_5$	
m_6	y_6A		m_6y_6
$\Delta=\sum$(Masses)		$\sum$(Moments A)	$\sum$(Moments F)

Table 6.2 – Calculation for LCG (Method A_

$$\mathrm{LCG_{MS}} = \frac{\sum(\text{Moments A}) - \sum(\text{Moments F})}{\sum(\text{masses})}$$

Masses	Lcg_{MS}	Moments
		Aft (+); Fwd (-)
Δ_L	$LCG_{LA}A$	$\Delta_L LCG_L$
m_1	y_1F	$-m_1y_1$
m_2	y_2A	m_2y_2
$-m_3$	y_3F	m_3y_3
m_4	y_4F	$-m_4y_4$
$-m_5$	y_5A	$-m_5y_5$
m_6	y_6A	$-m_6y_6$
$\Delta=\sum$(Masses)		$\sum$(Moments)

Table 6.3 – Calculation for LCG (Method B)_

$$\mathrm{LCG_{MS}} = \frac{\sum(\text{Moments})}{\sum(\text{masses})}$$

6.4.1.3 Determination of TCG_{CL}

If now moments are taken with reference to midship (MS), then by the vertical computation method, and considering aft moments to be positive and forward moments to be negative:

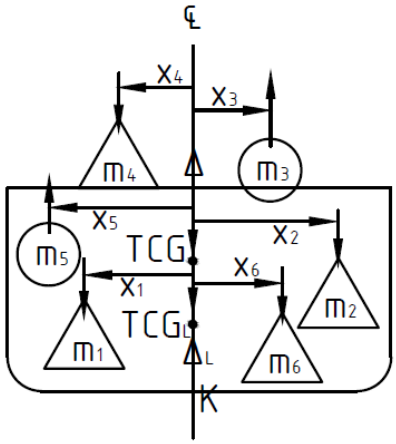


Figure 6.8 – Loaded TCG

Method A

Masses	Vcg_{CL}	Moments	
		Port (+)	Stbd (-)
Δ_L	VCG_{CL}	0	0
m_1	y_1P	m_1y_1	
m_2	y_2S		m_2y_2
$-m_3$	y_3P	$-m_3y_3$	
m_4	y_4S		m_4y_4
$-m_5$	y_5P	$-m_5y_5$	
m_6	y_6S		m_6y_6
$\Delta=\sum(\text{Masses})$		$\sum(\text{Moments P})$	$\sum(\text{Moments S})$

Table 6.4 – Calculation for TCG (Method A)

$$\text{VCG}_{\text{CL}} = \frac{\sum(\text{Moments P}) - \sum(\text{Moments S})}{\sum(\text{masses})}$$

Method B

Masses	Vcg_{MS}	Moments
		Port (+); Stbd (-)
Δ_L	0	0
m_1	y_1P	m_1y_1
m_2	y_2S	$-m_2y_2$
$-m_3$	y_3P	$-m_3y_3$
m_4	y_4S	$-m_4y_4$
$-m_5$	y_5P	$-m_5y_5$
m_6	y_6S	$-m_6y_6$
$\Delta=\sum(\text{Masses})$		$\sum(\text{Moments})$

Table 6.5 – Calculation for TCG (Method B)

$$\text{LCG}_{\text{MS}} = \frac{\sum(\text{Moments})}{\sum(\text{masses})}$$

6.4.1.4 Determination of KG, LCG_{MS} and TCG_{CL} when masses are shifted
When an intermediate position of the C of G for added and removed masses has been obtained as above, the results so obtained are then used as basis in a second exercise to calculate the effect of any shifted mass (m_S) to complete computation.